

ClipScore App Summary

Repo summary derived only from README, Python CLI/evaluator code, requirements, and local test data.

What It Is

A Python command-line toolkit for evaluating similarity between driving videos and text prompts with CLIP.

It supports single-video checks, scenario-level comparisons, and optional batch evaluation across local test data.

Who It's For

Primary user: a researcher or tester who needs to measure video-text alignment for driving scenario clips stored locally.

What It Does

- Loads OpenAI CLIP via Hugging Face `CLIPModel`` and `CLIPProcessor``.
- Extracts frames from local video files with OpenCV and converts them to PIL images.
- Computes per-frame cosine similarity scores between video frames and a text prompt.
- Aggregates mean, std, min, and max CLIP scores for each video.
- Calculates temporal consistency from adjacent-frame image embeddings.
- Runs quick tests in interactive mode or via CLI for scenarios, single videos, or scenario listing.
- Can evaluate multiple scenarios and save JSON results plus logs to a local `results/`` directory.

How It Works

- Entry points: `quick_clip_test.py`` exposes interactive and CLI flows; `video_clip_evaluator.py`` also has a batch-evaluation CLI.
- `QuickTester`` initializes `ImprovedVideoCLIP evaluator``, reads `test_data/description.txt``, and resolves scenario video files from local folders.
- `ImprovedVideoCLIP evaluator`` opens videos with OpenCV, samples frames, preprocesses them with `CLIPProcessor``, and runs inference with `CLIPModel`` on CPU or CUDA.
- The evaluator normalizes embeddings, computes cosine similarity per frame, summarizes scores, and separately computes adjacent-frame temporal consistency.
- Outputs stay local: console summaries by default, plus optional JSON and log files written under `results/`` by the batch evaluator.

How to Run

- From the repo root, install deps: `python3 -m pip install -r code/requirements.txt``.
- Run interactive mode: `python3 code/quick_clip_test.py``.
- Or run a single video test: `python3 code/quick_clip_test.py video --video-path "test_data/001376/crash.mp4" --text-prompt "...".``
- Batch CLI exists in `code/video_clip_evaluator.py``; local sample data lives under `test_data/``.